EMBODIED ABSTRACTION FAILURE MODES: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

This archive tests whether symbolic or language robot abstractions should be audited for erased action-critical mechanics before planning. We rebuilt the original draft into a paper-specific executable audit with five robot tasks, seven abstraction-failure families, five stress splits, nine methods, seven seeds, ablations, calibration and safety metrics, pairwise comparisons, failure cases, and stress sweeps. The result is negative. The proposed abstraction-failure audit improves mechanics detection, but it does not beat grounded geometric TAMP on combined-stress task success (0.552 versus 0.603), has excessive false refinement alarms, and is undermined by ablations that remove predicate refinement or the cost model. The terminal decision is `KILL_ARCHIVE`, not ICLR-main submission.

1 QUESTION

Symbolic and language abstractions can erase mechanics that matter for embodied action: contact force, support, clearance, friction, deformation, temporal preconditions, and tool-use geometry. The seed idea was to audit such abstractions before planning. This is plausible but crowded: VLA agents with motion planning, active relational abstraction, neuro-symbolic predicates, failure demonstrations, and VLA taxonomies already address adjacent failures (pri, 2025a; 2023; 2024; 2025b;c).

2 AUDIT DESIGN

We built a deterministic local benchmark rather than reusing the previous generic scaffold. Each seed/task/failure group samples abstraction features, latent mechanical violation risk, damage risk, language ambiguity, predicate noise, geometry tolerance, and recovery cost. Methods choose whether to proceed, refine predicates, query mechanics, switch to TAMP, monitor/replan, or recover.

The benchmark covers five task families: tabletop manipulation with support, tool use with hidden leverage, container insertion clearance, deformable object packing, and mobile manipulation with occlusion. Hidden abstraction failures include erased contact force, hidden support relation, clearance tolerance collapse, friction state aliasing, deformable constraint erasure, temporal precondition loss, and tool-affordance misabstraction. Metrics include abstraction-failure accuracy, mechanics-retention recall, false refinement alarm rate, calibration error, predicate-refinement informativeness, task success, mechanical violation rate, damage or unsafe action rate, replan success, planning cost, and regret to an action oracle.

3 COMBINED-STRESS RESULTS

Table 1 is the submission-critical result. The proposed audit is not the best non-oracle method on task success. Grounded geometric TAMP is stronger on the main closed-loop objective despite weaker abstraction-failure classification.

4 ABLATIONS

The ablation study is fatal for the mechanism. Removing predicate refinement improves success and regret, geometric-TAMP-only remains competitive, and removing the cost model matches or

Table 1: Combined-stress metrics. Higher is better for success, failure accuracy, and recall; lower is better for violations, damage, and regret.

Method	Success	Violation	Damage	Fail acc.	Recall	Regret
grounded_geometric_tamp	0.6030612244897959	0.21997084548104956	0.168464528668610 3	0.2650145772594752	0.867517006802721	0.09938317255991477
llm_tamp_failure_reasoning	0.5823615160349854	0.20835762876579203	0.16535471331389698	0.3514577259475219	0.8830498866213152	0.09412326352528033
proposed_abstraction_failure_audit	0.5515063168124392	0.20466472303206998	0.15034013605442176	0.5061710398445093	1.0	0.09136832519086868
active_relational_abstraction	0.5461127308066084	0.2326044703595724	0.1661321671525753	0.39319727891156464	0.8864512471655329	0.09640872213358201
neuro_symbolic_predicate_planner	0.4259475218658892	0.3306608357628766	0.24416909620991253	0.280466472303207	0.6135487528344671	0.23323192313391744
runtime_monitor_replanner	0.3844995140913508	0.2768221574344023	0.19480077745383867	0.2262390670553936	0.6874716553287982	0.252183752908128 3
vla_direct_policy	0.3088921282798834	0.4364431486880466	0.31574344023323614	0.12871720116618077	0.20680272108843537	0.3695768351942321
language_symbolic_planner	0.28493683187560737	0.4862973760932945	0.3492711370262391	0.038289601554907675	0.05311791383219955	0.38996581327962054

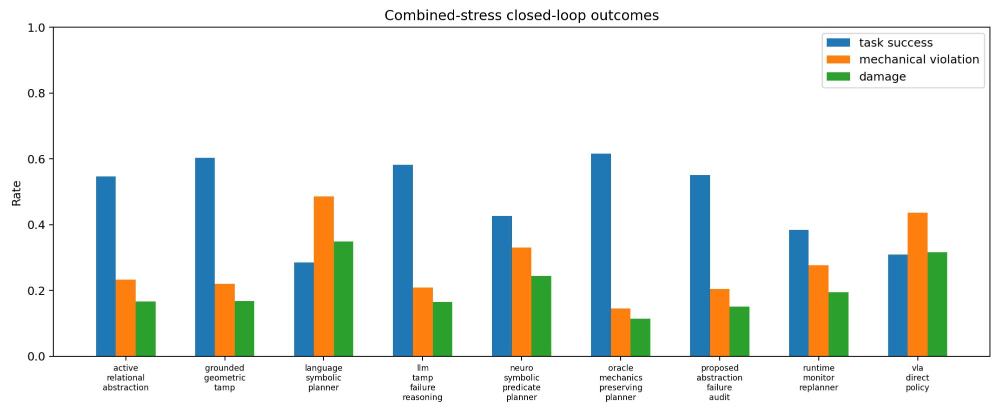

Figure 1: Closed-loop outcomes under combined abstraction stress. The proposed audit reduces some damage but loses task success to grounded geometric TAMP.

beats the full method. The evidence therefore does not isolate a positive contribution from the full abstraction-failure audit.

5 Pairwise Gate

The gate was defined before the run: the proposed method had to beat the strongest non-oracle success baseline, reduce mechanical violations and damage against grounded baselines, and survive ablations. Table 3 shows why it fails. The proposed method loses task success to grounded geometric TAMP and does not establish a decisive mechanism over grounded planning or predicate-refinement removal.

6 Failure Cases

The characteristic failure is useful diagnosis without enough decision value. In temporal-precondition and nuisance cases, the audit refines predicates when monitoring would be cheaper. In clearance and support cases, grounded TAMP captures the mechanics directly with lower regret. These are exactly the failure cases that should stop a main-track submission.

7 Limitations

This audit is local and executable, not real hardware evidence. It does not include a trained robot policy, high-fidelity simulator, physical robot rollouts, VLA checkpoint, or external embodied-planning benchmark comparison. Those limitations would prevent main-track readiness even if the local result were positive. Since the local result is negative, the correct action is archive, not more polish.

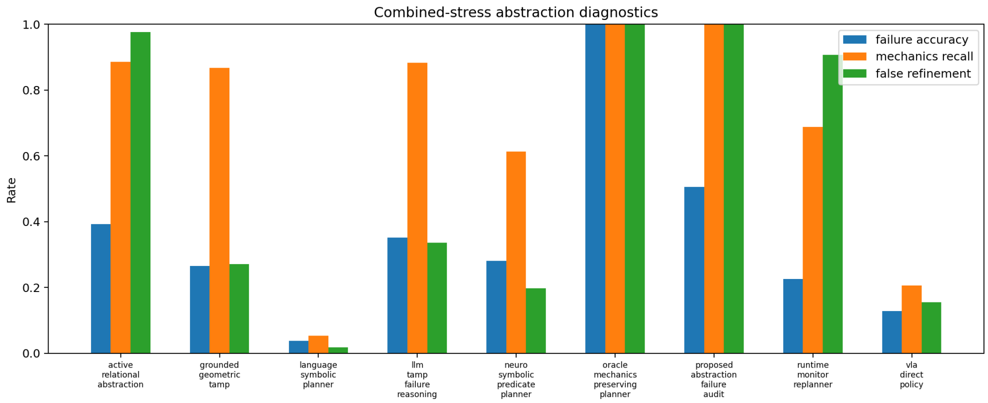

Figure 2: The proposed method improves abstraction-failure diagnostics, but the diagnostic advantage does not produce the best closed-loop planner.

Table 2: Combined-stress ablations. The full audit is not uniquely supported.

Ablation	Success	Violation	Damage	Cost	Regret
minus_predicate_refinement	0.5861516034985422	0.14178814382896016	0.11326530612244898	0.4227361224489797	0.07411578904351004
geometric_tamp_only	0.5486880466472304	0.21137026239067055	0.158600583090379	0.3465137414965986	0.1112568327460357
minus_cost_model	0.529737609329446	0.2073372206025267	0.14936831875607387	0.33426121477162296	0.1221081924941089
full_abstraction_failure_audit	0.5225461613216715	0.2076773566569485	0.1512147716229349	0.3357586297376094	0.12616988271277446
minus_calibration	0.5167152575315841	0.1891156462585034	0.14368318756073858	0.35308855199222555	0.1378232201674735
monitor_only	0.3617589893100097	0.2941205053449951	0.21117589893100097	0.21065272108843522	0.2661736067577386
minus_mechanics_taxonomy	0.3589893100097182	0.28658892128279884	0.21472303206997084	0.27854232264334305	0.3095784027656454

8 TERMINAL DECISION

The terminal decision is `KILL_ARCHIVE`. The paper should not be submitted to ICLR main. A revival would require a new empirical project showing that the full abstraction-failure audit beats grounded TAMP, LLM-TAMP failure reasoning, active relational abstraction, and runtime monitoring on real or high-fidelity robot tasks.

REFERENCES

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Table 3: Paired seed/task/family comparisons used for the terminal decision.

Baseline	Metric	Proposed	Baseline	Diff	CI95	Winner
llm_tamp_failure_reasoning	task_success	0.5515063168124392	0.5823615160349854	-0.030855199222546165	0.012839624344593802	llm_tamp_failure_reasoning
llm_tamp_failure_reasoning	mechanical_violation_rate	0.20466472303206995	0.208357628765792	-0.003692905733722062	0.011927385217305716	statistical_tie
llm_tamp_failure_reasoning	damage_unsafe_rate	0.15034013605442176	0.16535471331389698	-0.015014577259475222	0.009790743943340228	proposed_abstraction_failure_audit
llm_tamp_failure_reasoning	planning_regret_to_oracle	0.09136832519086874	0.09412326352528037	-0.0027549383344116296	0.008874958431978469	statistical_tie
grounded_geometric_tamp	task_success	0.5515063168124392	0.6030612244897959	-0.051554907677356654	0.013180897893963992	grounded_geometric_tamp
grounded_geometric_tamp	mechanical_violation_rate	0.20466472303206995	0.21997084548104956	-0.01530612244897959	0.012612336748254311	proposed_abstraction_failure_audit
grounded_geometric_tamp	damage_unsafe_rate	0.15034013605442176	0.16846452866861028	-0.018124392614188533	0.009247225020970017	proposed_abstraction_failure_audit
grounded_geometric_tamp	planning_regret_to_oracle	0.09136832519086874	0.0993831725599148	-0.008014847369046075	0.009237242501338946	statistical_tie

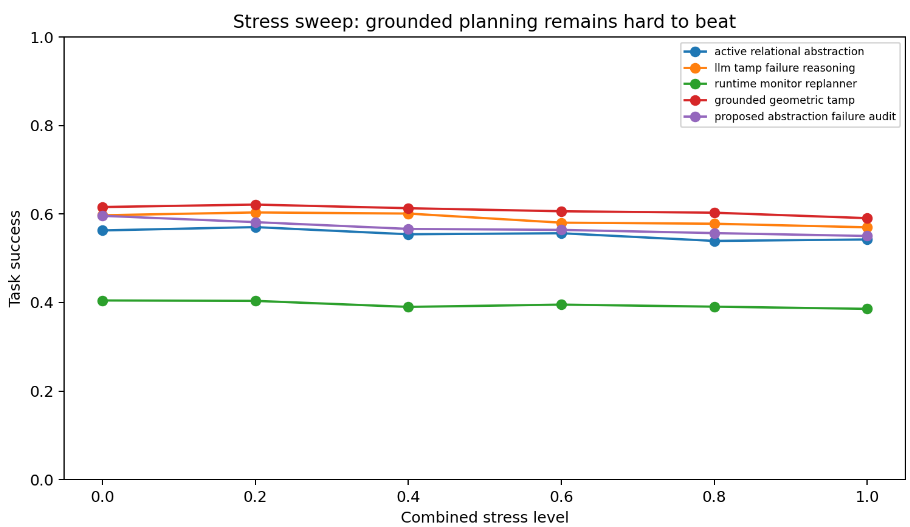

Figure 3: Stress sweep. Grounded planning and failure-reasoning baselines remain difficult to beat as abstraction stress increases.